

VasanthaKumar A

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SUMMARY

Results-driven Artificial Intelligence and Robotics student with hands-on experience in robotics simulation, intelligent automation, computer vision, AI systems, and real-world deployment. Experienced in leading technical teams, developing AI-powered applications, conducting workshops, and building scalable solutions using modern AI/ML technologies.

EXPERIENCE

Robotics Engineer Intern

Dec 2025 – Present

Zentron Labs, Trichy, Tamil Nadu

- Built a computer vision segmentation pipeline using SAM2 for dataset creation and trained models using U-Net.
- Configured and deployed end-to-end robot simulation environments using Gazebo and ROS2.
- Designed and implemented a steering control system for a 3-wheeled robotic vehicle, validating turning behavior and stability through simulation testing. [\[Demo\]](#)
- Built a web-based UI for robotic arm simulation, enabling interactive joint-level control and real-time motion validation. [\[Demo\]](#)
- Integrated simulation environments with real-world robotic behavior through iterative testing and debugging.
- Developed a URDF editor featuring 3D visualization, Xacro conversion, and direct simulation deployment.

Team Lead – AI & Robotics Initiatives

2025

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- Spearheaded a 30+ member team in developing AI-driven applications, robotics solutions, and intelligent automation systems.
- Built and scaled projects leveraging AI/ML, agentic AI, and automation technologies for real-world applications.
- Organized and delivered AI & Robotics workshops, technical campaigns, and outreach programs impacting 500+ students.
- Mentored participants and evaluated innovative projects during technical events including UTSAV'25 Hackathon.
- Created AI-focused educational content, building a community of 3,000+ followers while promoting AI literacy, robotics, and emerging technologies. [\[Instagram\]](#)

Anukul Shiksha Trainer

2025

SASTRA University, Thanjavur

- Conducted intensive pre-placement training sessions focused on aptitude, logical reasoning, communication, and interview preparation.
- Trained and mentored 200+ students to improve analytical thinking and career readiness.

PUBLICATIONS

- **Unified Real-Time Compliance Monitoring System** – Accepted for publication at **ICAIES-2025** (Springer LNCS Series, SCOPUS Indexed). Ranked among the Top 20% of submissions and presented at an international conference.

PROJECTS

Sana-V: Deep RL-Powered Assistive Robot

2025

- Built a 4-wheeled robot capable of following users, speech interaction, and behavior monitoring using camera-based AI perception and Deep Reinforcement Learning.
- Designed an ADHD monitoring prototype with person tracking, autonomous patrols, and mobile-controlled accessibility features.
- **Technologies:** ESP32, ESP32-CAM, Ultrasonic Sensors, Python, Deep Reinforcement Learning.

Unified Real-Time Compliance Monitoring System2025

- Implemented a real-time traffic monitoring system to detect helmet violations, speeding, overloading, facial recognition, and license plate extraction.
- **Technologies:** Python, OpenCV, Deep Learning, Detection & Tracking, Computer Vision.

3D Design Customizer Web App2025

- Developing an AI-driven platform for customized product visualization using Three.js.
- Enabled automatic placement of logos, text, and images on 3D product models including bags, boxes, and bottles.
- **Technologies:** JavaScript, Three.js, GANs, Python.

Recreating Google MediaPipe Pose Detection System2024

- Built an advanced pose estimation model using the COCO 2017 dataset.
- Implemented residual blocks and attention mechanisms for improved pose estimation accuracy.
- **Technologies:** Python, OpenCV, PyTorch.

EDUCATION

SASTRA University2022 – Present

Bachelor of Technology in Artificial Intelligence and Robotics

- Specialization in AI/ML, Computer Vision, and Multi-Agent Systems.
- Active participation in robotics research and development projects.

ANNS Matriculation and Higher Secondary School2021

TECHNICAL SKILLS

Programming Languages: Python, C, C++

AI/ML & Automation: CNNs, LLMs, GANs, LSTMs, DRL, NLP, RAG, AI Agents, Computer Vision, ComfyUI, n8n

Web & Application Development: FastAPI, Streamlit, Flutter, Three.js

Simulation & Robotics: Gazebo, ROS2, MATLAB, CoppeliaSim, Raspberry Pi, ESP32, ESP32-CAM, Arduino IDE, Jetson Nano, 3D Printing, AI CCTV Pipelines

Leadership & Business: Team Management, Startup Scaling, Technical Mentoring, Marketing Strategies, Educational Campaigns

LANGUAGES

Tamil (Native), English (Fluent)